Supplementary Material: Text-Guided Diffusion for Ambiguity-Aware Medical Image Segmentation

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ABSTRACT

This supplementary material accompanies the ICVGIP2026 paper AmbiguousTextDiff. It collects the training and inference pseudocode, the full hyperparameter table, a compact architecture schematic, additional qualitative results for LIDC-IDRI and IMA++, visual analyses of the ablations, 16-sample ensembles, semantic-control tests, sample-count sweep, and KL-weight sweep, baseline training details, and a pointer to the code repository.

ACM Reference Format:
Venkata Siddharth Dhara, Pawan Kumar, and Pratik Jawanpuria. 2026. Supplementary Material: Text-Guided Diffusion for Ambiguity-Aware Medical Image Segmentation. In *Proceedings of 17th Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP2026).* ACM, New York, NY, USA, 6 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 TRAINING AND INFERENCE PSEUDOCODE

Algorithm 1 and Algorithm 2 give pseudocode for the ambiguity-aware training objective and the diverse-ensemble inference procedure described in Sections 3.3 and 3.4 of the main paper.

Algorithm 1 AmbiguousTextDiff training step

Require: image $\boldsymbol{x}$, annotator mask $\boldsymbol{y}_0$, text prompt $\boldsymbol{c}$, diffusion steps T, KL weight λ

Sample $t \sim \text{Uniform}(1, T)$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
$\boldsymbol{y}_t \leftarrow \sqrt{\bar{\alpha}_t}\, \boldsymbol{y}_0 + \sqrt{1-\bar{\alpha}_t}\, \boldsymbol{\epsilon}$ ▷ forward noising
$\boldsymbol{z}_t \leftarrow [\boldsymbol{x}, \boldsymbol{y}_t]$ ▷ channel-wise image fusion
$\boldsymbol{\epsilon}_{\boldsymbol{\theta}} \leftarrow \text{UNet}(\boldsymbol{z}_t, \boldsymbol{c}, t)$ ▷ text fused via cross-attention
$\hat{\boldsymbol{y}}_0 \leftarrow (\boldsymbol{y}_t - \sqrt{1-\bar{\alpha}_t}\, \boldsymbol{\epsilon}_{\boldsymbol{\theta}})/\sqrt{\bar{\alpha}_t}$
$q \leftarrow q_{\boldsymbol{\phi}}(\boldsymbol{u} \mid \boldsymbol{x}, \boldsymbol{y}_0)$; $\quad p \leftarrow p_{\boldsymbol{\psi}}(\boldsymbol{u} \mid \boldsymbol{x}, \hat{\boldsymbol{y}}_0)$
$\mathcal{L}_{\text{mse}} \leftarrow \|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_{\boldsymbol{\theta}}\|^2$; $\quad \mathcal{L}_{\text{kl}} \leftarrow D_{\text{KL}}(q \,\|\, p)$
$\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{mse}} + \lambda\, \mathcal{L}_{\text{kl}}$
Backpropagate $\mathcal{L}_{\text{total}}$; update $\boldsymbol{\theta}$ with AdamW; update EMA weights

2 CONSOLIDATED HYPERPARAMETERS

Table 1 gathers the hyperparameters described in Section 4.2 of the main paper so that the training and inference setup can be reproduced directly.

Algorithm 2 Diverse ensemble sampling

Require: image $\boldsymbol{x}$, fixed case-level text prompt $\boldsymbol{c}$, ensemble size $N = 16$

for $i = 1$ **to** N **do**
  Sample $\boldsymbol{y}_T^{(i)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
  for $t = T$ **down to** 1 **do**
    $\boldsymbol{\epsilon}_{\boldsymbol{\theta}} \leftarrow \text{UNet}([\boldsymbol{x}, \boldsymbol{y}_t^{(i)}], \boldsymbol{c}, t)$
    $\boldsymbol{y}_{t-1}^{(i)} \leftarrow \text{DDPM/DDIMStep}(\boldsymbol{y}_t^{(i)}, \boldsymbol{\epsilon}_{\boldsymbol{\theta}}, t)$
  end for
  $\boldsymbol{y}^{(i)} \leftarrow \mathbb{1}[\boldsymbol{y}_0^{(i)} \geq 0.5]$
end for
$\bar{\boldsymbol{y}} \leftarrow \frac{1}{N} \sum_i \boldsymbol{y}^{(i)}$
$\boldsymbol{y}_{\text{maj}} \leftarrow \mathbb{1}[\bar{\boldsymbol{y}} \geq 0.5]$
return $\{\boldsymbol{y}^{(1)}, \ldots, \boldsymbol{y}^{(N)}\}, \bar{\boldsymbol{y}}, \boldsymbol{y}_{\text{maj}}$

Table 1: AmbiguousTextDiff hyperparameters.

Hyperparameter	Value
Diffusion steps T	1000
Noise schedule	linear, β_1=10^{-4}, β_T=0.02
U-Net base width	128
Attention resolutions	32, 16, 8
Attention head width	64 channels/head
Text encoder	Bio_ClinicalBERT (frozen), 768-d → 128-d
Text max. length	128 tokens (longer prompts truncated)
Optimizer	AdamW, no weight decay
Learning rate	10^{-4}, linear anneal to 0
Batch size	4
Gradient clipping	max norm 1.0
EMA decay	0.9999
KL weight λ	10^{-3}
Training steps	50,000
Inference samples N	16
Inference sampler	DDPM (1000 steps) or DDIM (100 steps)
Image resolution	128×128, normalized to $[-1, 1]$
Hardware	4×RTX 4090 GPUs

Evaluation note. The same settings are used for all quantitative tables in the main paper. Each stochastic method is scored with the same number of generated masks and the same binarization convention, so GED, CI, $D_{\max}$, S_c, and DA reflect the learned mask distribution rather than a different evaluation protocol.

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ICVGIP2026, December 21-25, 2026, Kolkata, India

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ACM ISBN 978-x-xxxx-xxxx-x/YY/MM.
https://doi.org/10.1145/nnnnnnn.nnnnnnn

3 ARCHITECTURE SCHEMATIC

Figure 1 gives the compact denoiser-level schematic corresponding to the architecture described in Section 3.2 of the main paper. The figure highlights that the image and noisy mask enter by channel-wise concatenation, while the frozen text encoder provides projected key/value features to cross-attention blocks at multiple U-Net resolutions.

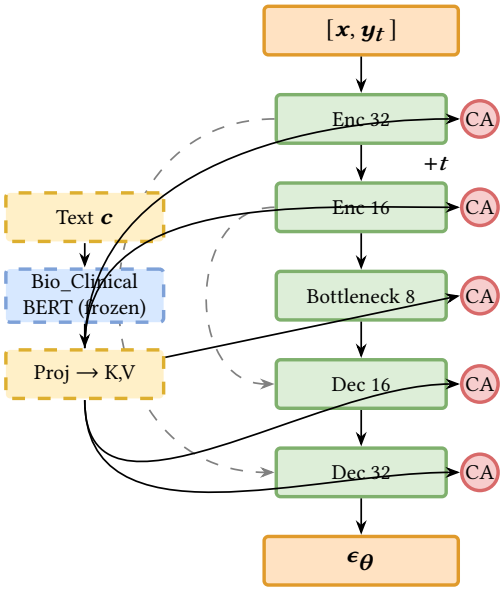


Figure 1: Compact denoiser schematic. The image+noisy-mask tensor is fused by channel-wise concatenation, while Bio_ClinicalBERT text features are projected into key/value vectors and injected through cross-attention blocks at the 32/16/8 U-Net resolutions.

4 SUPPLEMENTARY QUALITATIVE RESULTS

Figures 2, 3, 4, 5, and 6 provide the visual evidence supporting the qualitative discussion in the main paper. The sequence mirrors the experimental narrative: cross-method comparisons first, then component ablations, full-ensemble behavior, and semantic-control stress tests. These panels are included here to make boundary detail, empty-mask handling, artifact robustness, component behavior, and prompt sensitivity easier to inspect.

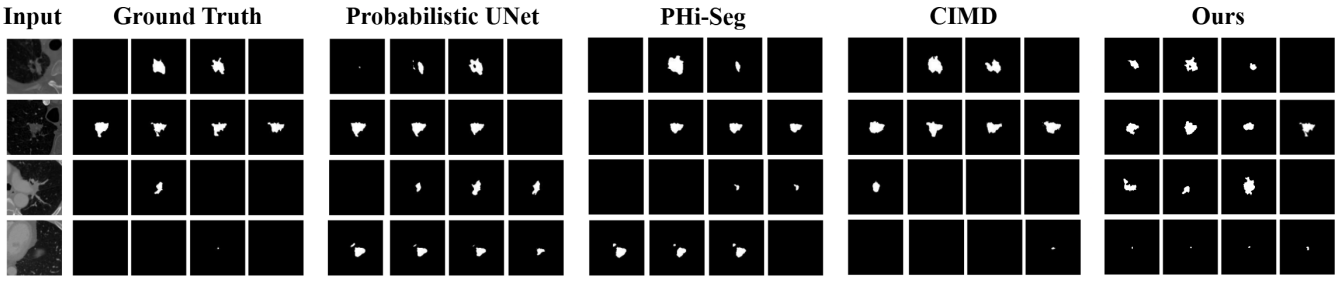


Figure 2: Qualitative LIDC-IDRI comparison. Four expert annotations and the first four samples from each stochastic method are shown, including cases where the ground truth contains empty masks.

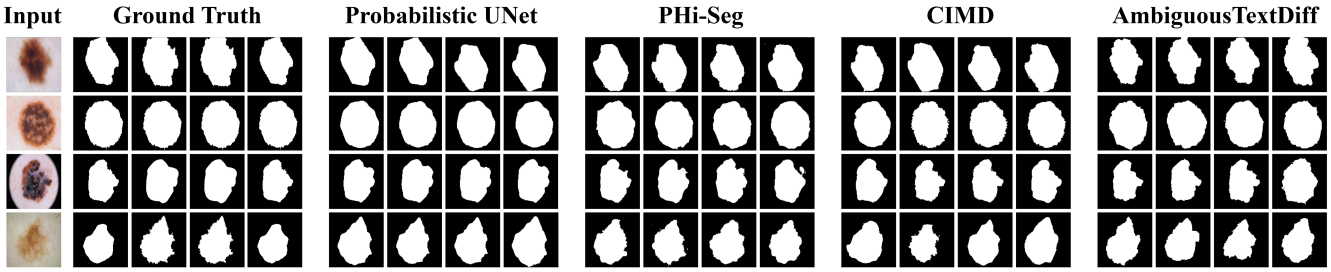


Figure 3: Qualitative comparison on IMA++. For each case (left), four annotator masks are shown alongside the first four samples from Probabilistic U-Net, PHiSeg, CIMD, and AmbiguousTextDiff.

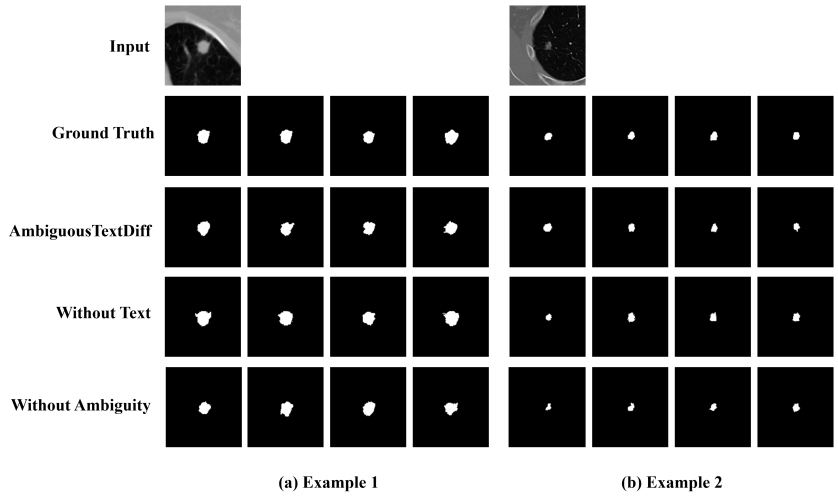


Figure 4: Qualitative ablation on LIDC-IDRI. For two cases, we show the input CT slice, four expert ground-truth annotations, and the first four samples from the full model, the w/o-Text-Guidance variant, and the w/o-Ambiguity-Modeling variant. The no-text variant is less shape-consistent, while removing the KL term narrows the samples toward a single mode.

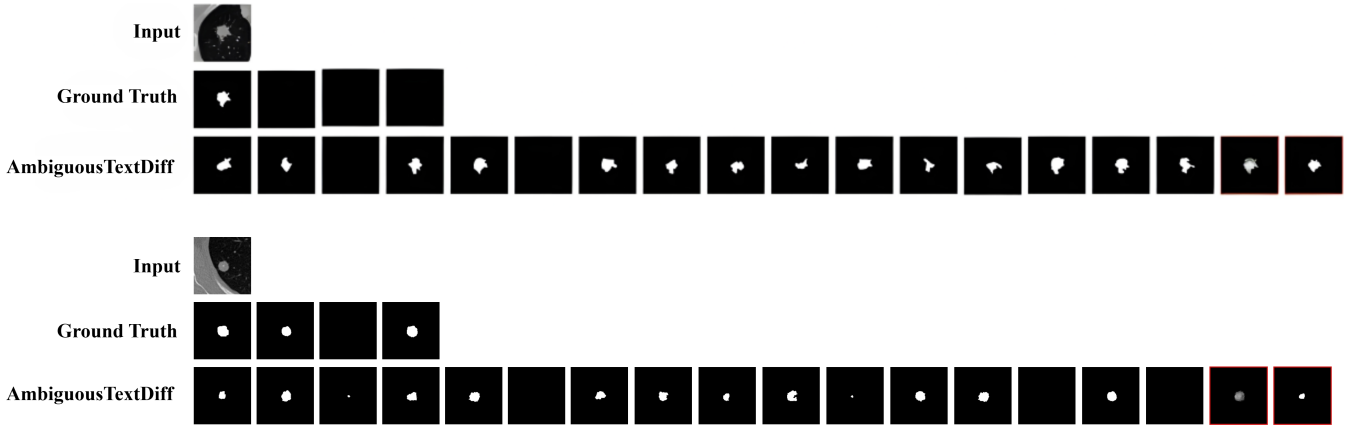


Figure 5: Full 16-sample ensemble for two representative LIDC-IDRI cases. The ensemble average and majority-vote summaries at the end of each row provide a compact uncertainty view derived directly from the generated samples.

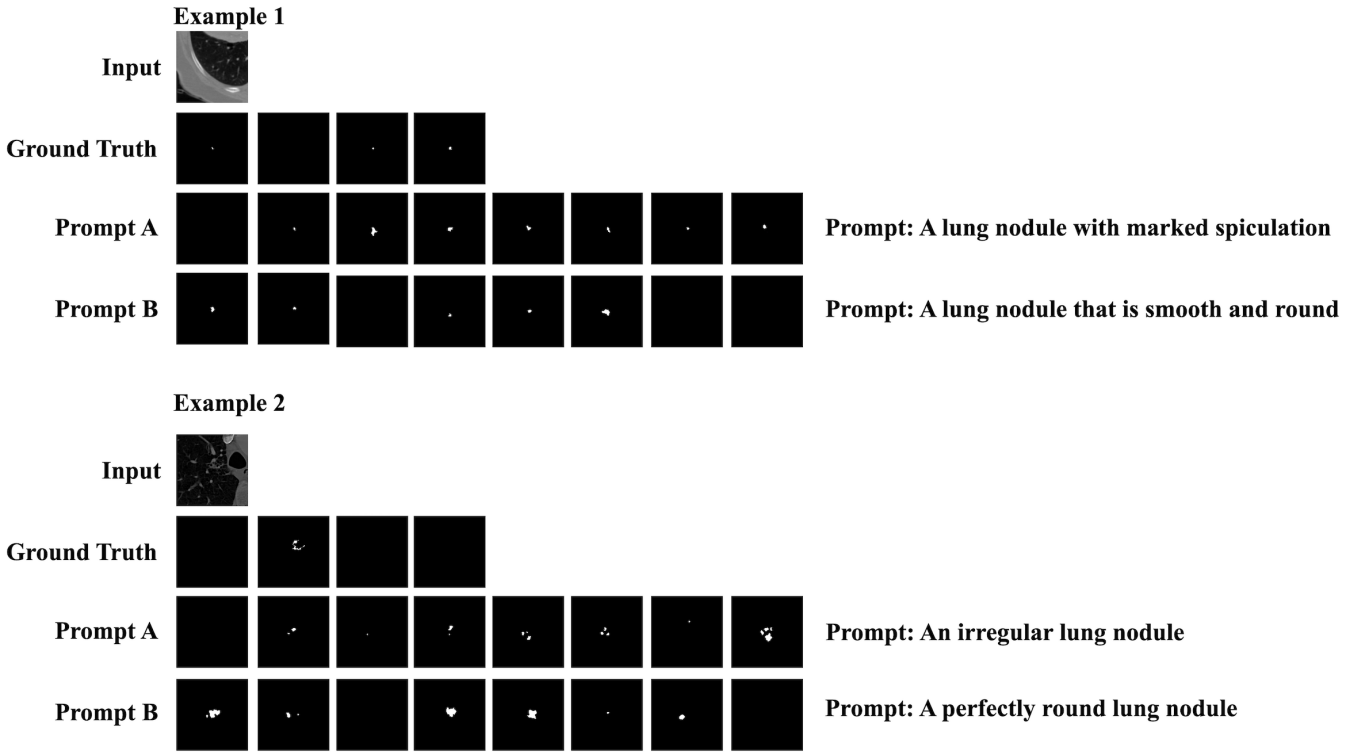


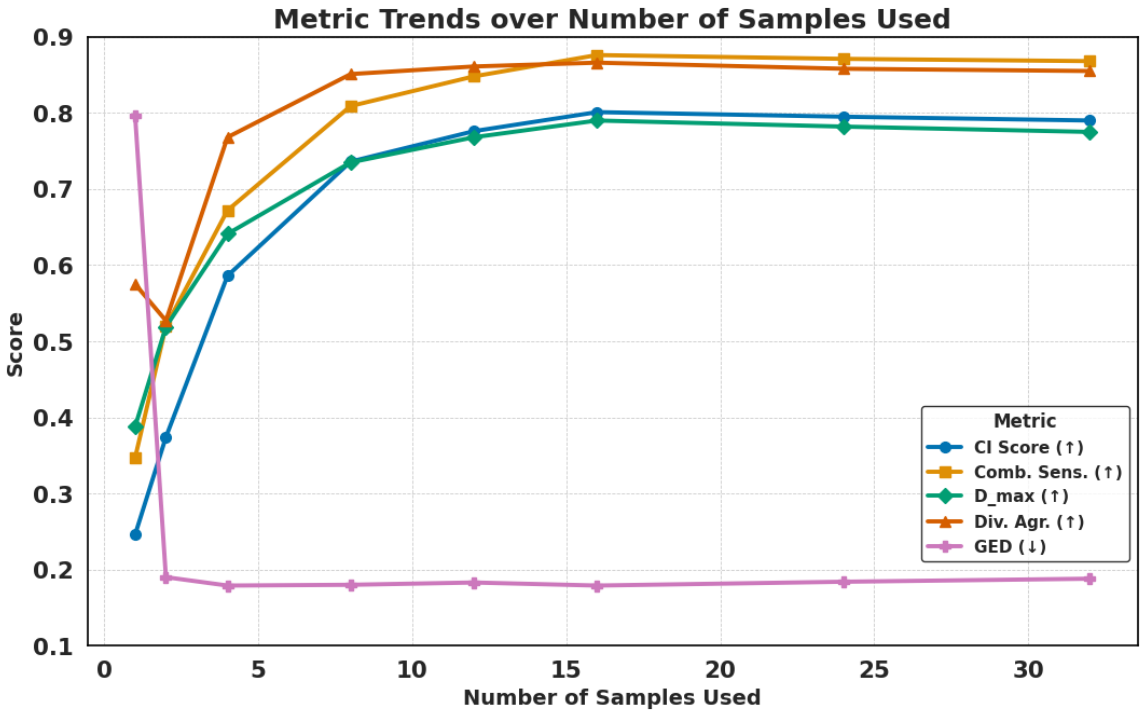
Figure 6: Semantic-control stress test. For the same input image, changing the text from a factual descriptor to a contradictory descriptor shifts generated masks in the expected direction, while the standard inference protocol in the main paper keeps a single case-level prompt fixed across all samples.

Interpretation of qualitative panels. The supplementary qualitative panels separate three effects that are summarized more compactly in the main paper. Figures 2 and 3 show that the method remains close to the annotator-supported mask set across two visually different modalities: lung CT cases contain frequent present/absent nodule ambiguity, whereas IMA++ cases mainly test boundary variability and artifact robustness in dermoscopy. In both settings, the generated samples are not intended to be identical replicas of a single annotation; the target behavior is to cover the plausible region spanned by multiple annotators while avoiding masks unsupported by the image.

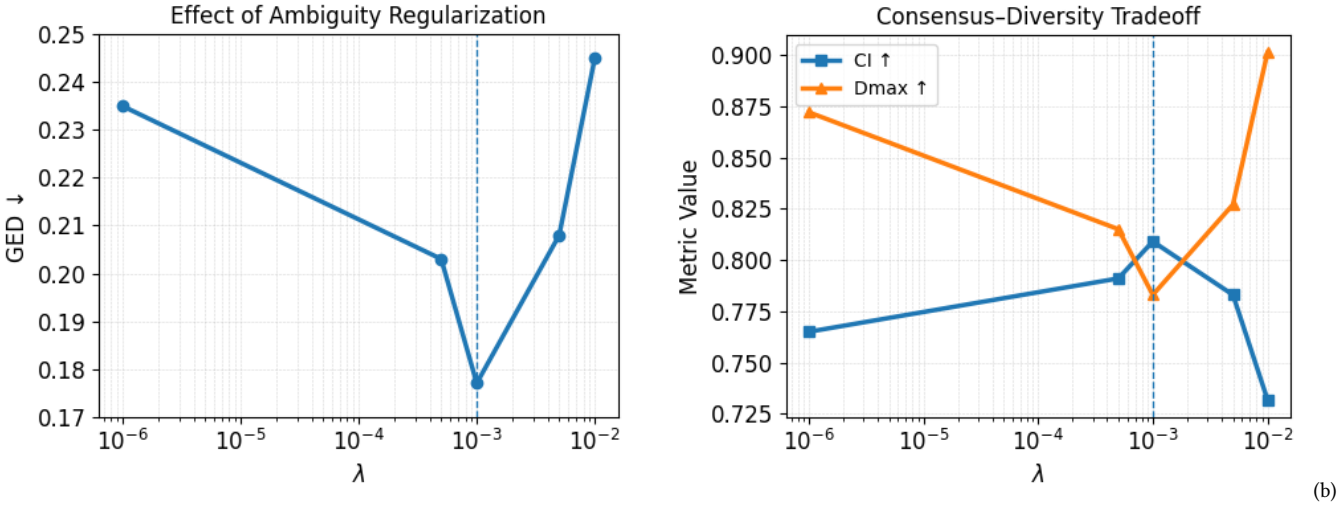
Figure 4 isolates the two components of this behavior. Removing text guidance keeps stochastic sampling but weakens the consistency between shape and descriptor-level context, while removing ambiguity modeling narrows the generated set toward fewer modes. Figure 5 then shows the complete N=16 ensemble used for evaluation, making clear that the ensemble average and majority vote are computed from the same samples rather than from a separate post-processing model. Finally, Figure 6 is a controlled prompt-sensitivity test: standard inference keeps one factual case-level prompt fixed for all samples, but deliberately changing the descriptor for the same image shifts the generated masks in the expected direction, confirming that cross-attention provides usable semantic conditioning.

5 ADDITIONAL ANALYSIS FIGURE

Figure 7 expands the analysis plots summarized in the main paper. It is included here to make the chosen ensemble size and KL-weight setting easier to inspect. The sample-count plot shows that most distributional metrics improve rapidly from one to sixteen generated masks, after which the curves are nearly saturated; this supports using N=16 as a practical balance between coverage and inference cost. The KL-weight plots complement the ablation study by showing that ambiguity modeling is not an all-or-nothing switch: moderate regularization gives the best GED/CI behavior, while too little or too much regularization moves the generated distribution away from the annotator-supported set. This analysis also links the implementation choices in Table 1 to the qualitative behavior in Figures 2-6: the sampling budget determines how much of the annotator distribution can be covered, while λ controls whether those samples remain diverse without drifting away from image-supported masks.



(a) Number of generated samples.



(b) KL-weight sweep.

Figure 7: Additional analysis plots. Left: metrics improve sharply up to N=16 and then saturate or slightly regress, motivating the fixed N=16 ensemble. Right: earlier single-seed λ sweep (DDIM); the retrained sweep in the main paper supersedes it and shows a broad stable regime for $\lambda \leq 10^{-3}$.

Together, these plots connect the qualitative behavior in Figures 4-6 to the numerical settings used in the main experiments. Increasing the ensemble size primarily improves distributional coverage until the generated set already spans the dominant annotator modes, while the retrained KL sweep shows a broad stable regime for $\lambda \leq 10^{-3}$ and degradation at 10^{-2}. The reported setting uses the same fixed N=16 and λ=10^{-3} across datasets rather than tuning them per example.

6 BASELINE TRAINING PROVENANCE

All stochastic baselines were retrained on our split with the same metric code and N=16 protocol (native 128×128 crops).

- Probabilistic U-Net: Adam, LR 10^{-4}, batch 32, 240k iterations, latent dim 6, β=1.
- PHiSeg: Adam, LR 10^{-3}, batch 12, 100 epochs, hierarchical prior.
- Generalized Probabilistic U-Net: Adam, LR 10^{-4}, batch 32, early stopping, minimum-validation-loss checkpoint.
- CIMD: Adam, LR 10^{-4}, batch 16, 50k steps, EMA 0.9999, T=1000, full DDPM sampling.
- AmbiguousTextDiff: AdamW, LR 10^{-4}, batch 4, 50k steps, EMA 0.9999, gradient clipping 1.0, T=1000.

Deviations from the original recipes include 128×128 rather than 180×180 inputs for the Probabilistic U-Net variants.